

Lecture 3B - Causal Mediation Analysis

Applying Potential Outcomes Framework to Mediation

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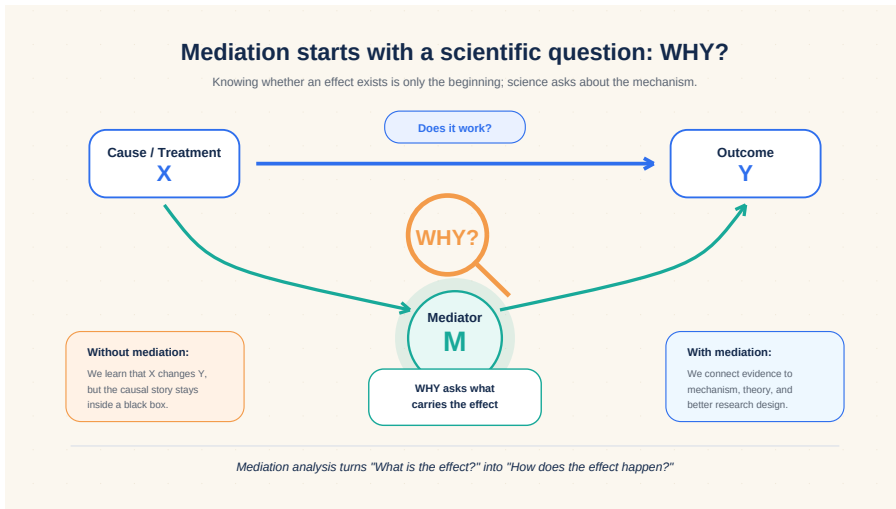
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- 1 Causal mediation analysis: basic setup
- 2 Causal mediation analysis: identification
- 3 Potential workarounds for the full-on mediator manipulation

Mediation is important



Common mediation practices

Baron and Kenny (1986) approach

Regression checklist

- 1 Show total effect: $D \rightarrow Y$
- 2 Show treatment predicts mediator:
 $D \rightarrow M$
- 3 Regress Y on D and M ; M should predict Y
- 4 Mediation inferred when the $D \rightarrow Y$ coefficient is reduced

What it gives:

A simple path-analysis recipe.

Bootstrapping indirect effect (Preacher and Hayes, 2008)

Resampling the product path

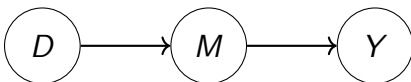
- Estimate the indirect effect: $a \times b$
- Resample cases to form a bootstrap confidence interval
- Declare mediation when the interval excludes zero

What it improves:

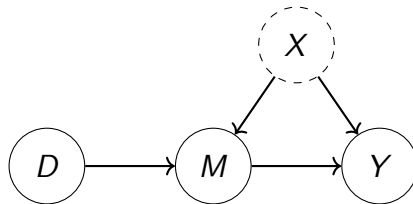
Inference for a non-normal indirect effect.

The central challenge in mediation analysis

Ideal case: mediator unconfounded



Common problem: mediator confounded



Key message

We are not okay with D being non-randomized, but we often seem comfortable treating M as if it were unconfounded.

① Causal mediation analysis: basic setup

② Causal mediation analysis: identification

③ Potential workarounds for the full-on mediator manipulation

From statistical paths to causal questions

Mediation asks a scientific “why” question:

- Statistical mediation: is the product path $a \times b$ nonzero?
- Causal mediation: what potential outcomes would change if the mediator changed?

Transition

The potential outcome framework helps us define the causal objects before estimating them.

Basic setup

Consider the simple case:

- Binary treatment: $D_i \in \{0, 1\}$
- Continuous mediator: M_i
- Continuous outcome: Y_i

$$D_i \longrightarrow M_i \longrightarrow Y_i$$

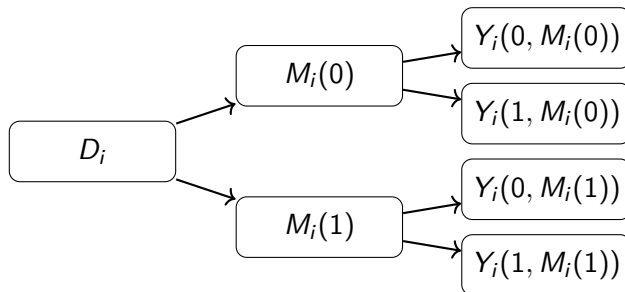
We distinguish two kinds of potential outcomes:

$$M_i(d) \quad \text{and} \quad Y_i(d, m)$$

Potential mediators and potential outcomes

$M_i(d)$: mediator value if treatment were set to d

$Y_i(d, m)$: outcome if treatment were set to d and mediator to m



Total effect

The total effect compares the outcome under treatment versus control:

$$TE_i = Y_i(1, M_i(1)) - Y_i(0, M_i(0)).$$

This is the overall effect of changing D from 0 to 1.

Key point

The total effect allows both pathways: D may affect Y directly and through M .

Controlled direct effect

The **controlled direct effect** holds the **mediator fixed at a chosen value m** :

$$CDE_i(m) = Y_i(1, m) - Y_i(0, m).$$

What is the effect of D if we intervene to keep the mediator at m ?

Interpretation

This is a direct effect under a controlled mediator intervention.

Natural direct effect

The **natural direct effect** holds the mediator at the level it would take under control:

$$NDE_i = Y_i(1, M_i(0)) - Y_i(0, M_i(0)).$$

What is the effect of changing D , while keeping M at its natural control-world value?

$$D : 0 \rightarrow 1, \quad M : M_i(0) \text{ fixed.}$$

Natural indirect effect

The **natural indirect effect** changes the mediator from its control value to its treatment value, while holding treatment fixed:

$$NIE_i(d) = Y_i(d, M_i(1)) - Y_i(d, M_i(0)).$$

The expectation of NIE or **the average causal mediation effect (ACME)**:

$$ACME(d) = \mathbb{E}[Y_i(d, M_i(1)) - Y_i(d, M_i(0))].$$

Interpretation

ACME is the portion of the treatment effect transmitted through the mediator.

Decomposing the total effect

One decomposition is:

$$TE_i = \underbrace{Y_i(1, M_i(1)) - Y_i(1, M_i(0))}_{ACME \text{ under } D=1} + \underbrace{Y_i(1, M_i(0)) - Y_i(0, M_i(0))}_{NDE}$$

Equivalently:

$$TE = ACME(1) + NDE(0).$$

Important

The decomposition is written in terms of counterfactual outcomes, not regression coefficients, so you know what you need for ACME and TE.

Key insights from causal mediation analysis

Causal mediation analysis (Imai et al., 2010) is useful because it makes the mediation question explicit:

- What would M_i have been under treatment versus control?
- What would Y_i have been if we changed D but held M fixed?
- What would Y_i have been if we changed M but held D fixed?

Takeaway

The potential outcome framework turns mediation from a path diagram into a set of well-defined counterfactual causal effects.

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Identification: what do we need to know?

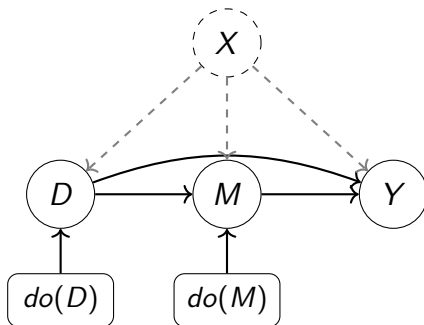
Recall the causal mediation estimand: $ACME(d) = \mathbb{E}[Y_i(d, M_i(1)) - Y_i(d, M_i(0))]$.

- ① How treatment changes the mediator: $M_i(1) - M_i(0)$.
- ② How changes in the mediator affect the outcome: $Y_i(d, m') - Y_i(d, m)$.

Key issue

Randomizing D helps identify the effect of D on M , but it does not automatically identify the effect of M on Y .

The ideal DAG: intervene on both D and M



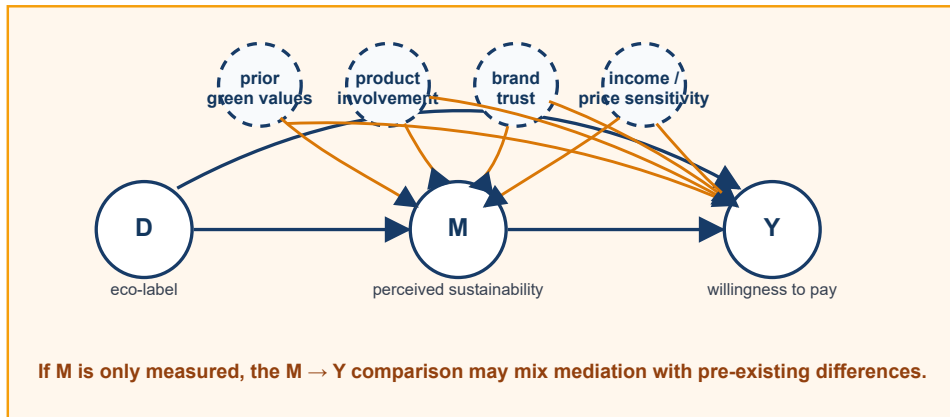
Interventions break the dependence of D and M on confounder X .

Key experimental condition

If both D and M may be confounded, identification requires design or assumptions that make both behave as if randomized.

Example: Eco-label and willingness to pay

Suppose we study whether an eco-label increases willingness to pay.



What needs to be learned from the data?

Mediator model

How does treatment change the mediator?

$$\mathbb{E}[M_i \mid D_i = d].$$

This tells us how the distribution of $M_i(d)$ changes with d .

Outcome model

How does the outcome depend on treatment and mediator?

$$\mathbb{E}[Y_i \mid D_i = d, M_i = m].$$

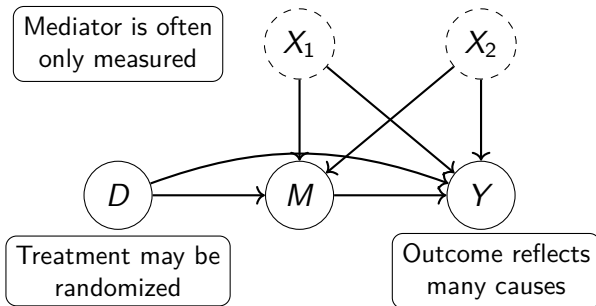
This tells us how $Y_i(d, m)$ changes with m .

Identification requirement

These observed-data relationships are causal only if the treatment and mediator are unconfounded, usually by design or strong assumptions.

Why mediator identification is hard in practice

In many studies, D is randomized but M is only measured (or hard to manipulate).



Conclusion

If we only measure M , the counterfactual terms in ACME are not identified unless we can rule out mediator–outcome confounding.

Estimation when both D and M are manipulated

ACME can be estimated via the following steps:

- 1 For each treatment condition $d \in \{0, 1\}$ and mediator value m :

$$\mu(d, m) = \mathbb{E}[Y_i \mid do(D = d), do(M = m)].$$

- 2 The mediator distribution under each treatment is also learned experimentally:

$$F_{M(d)}(m) = \Pr(M_i \leq m \mid do(D = d)).$$

- 3 Then the ACME can be written as:

$$ACME(d) = \int \mu(d, m) dF_{M(1)}(m) - \int \mu(d, m) dF_{M(0)}(m).$$

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 - More feasible (but lesser) designs

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When M is only measured: sensitivity analysis

The key concern is unobserved mediator–outcome confounding:

$$M_i \leftarrow U_i \rightarrow Y_i.$$

Sensitivity analysis asks:

How strong would the unobserved confounding have to be to make the estimated mediation effect disappear?

Logic

This is an *abductio ad absurdum* argument: assume the mediation result is not causal, then ask whether the required confounding is plausible.

Sensitivity analysis in a standard linear setup

A common mediation model is:

$$M_i = \alpha_0 + \alpha_1 D_i + \alpha_2 X_i + e_{Mi}$$

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 M_i + \beta_3 X_i + e_{Yi}$$

Full identification implies:

$$\text{Corr}(e_{Mi}, e_{Yi}) = 0.$$

Sensitivity analysis relaxes this:

$$\rho = \text{Corr}(e_{Mi}, e_{Yi}).$$

How to interpret sensitivity results

Key sensitivity parameter

$$\rho = \text{Corr}(e_M, e_Y)$$

- $\rho = 0$: no residual mediator–outcome confounding
- $\rho \neq 0$: unobserved confounding

Robustness question

- Recompute ACME over possible values of ρ
- Find the value where ACME becomes zero
- Judge whether that level of confounding is plausible

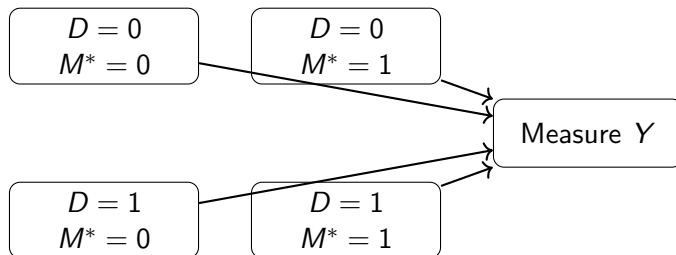
Reporting

Do not just report “significant mediation.” Report how sensitive the ACME is to mediator–outcome confounding.

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Ideal design: concurrent double randomization

Randomize both the treatment and the mediator manipulation in the same study.



Key idea

This directly estimates how manipulated mediator states affect the outcome under different treatment conditions.

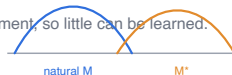
The difficulty of having the ideal design

However, fully manipulating M can be difficult.

1. How much should M be shifted?

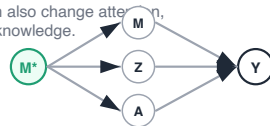
Too strong: no overlap and a different target estimand.

Too weak: no movement, so little can be learned.



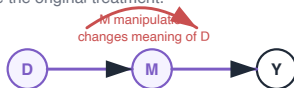
2. It may change other mechanisms

The M manipulation can also change attention, trust, beliefs, mood, or knowledge.



3. It may interfere with D

A separate M manipulation can overwrite, dilute, or redefine the original treatment.



4. Ethical or practical constraints

Some mediators are inappropriate, costly, slow, or impossible to induce experimentally.



Design-based solution 1: experimental causal chain

Instead of estimating the full indirect effect in one study, test the two causal links separately.

Study 1: Does D change M ?



Randomize eco-label. Measure perceived sustainability.

Study 2: Does M change Y ?



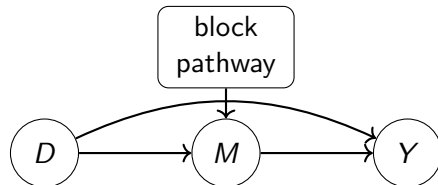
Manipulate sustainability perception. Measure willingness to pay.

Strength and limitation

Good for causal evidence about the chain, but it does not automatically estimate the exact ACME.

Design-based solution 2: blockage design

Block the proposed mediator pathway and see whether the treatment effect shrinks.



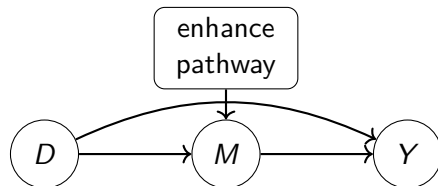
Example: if an eco-label works through sustainability beliefs, provide all participants with clear sustainability information so the label no longer changes that belief.

Interpretation

If the treatment effect is reduced when the mediator pathway is blocked, that supports the proposed mechanism.

Design-based solution 3: enhancement design

Amplify the proposed mediator pathway and see whether the treatment effect grows.



Example: pair the eco-label with additional sustainability cues to strengthen the perceived-sustainability pathway.

Interpretation

If the treatment effect increases when the pathway is enhanced, that supports the mediator as part of the causal process.

Where this leaves us

- Sensitivity analysis keeps the original mediation model, but asks how fragile it is.
- Experimental causal chain designs test the two causal links separately.
- Blockage and enhancement designs test whether the proposed pathway matters.

Main takeaway

When M cannot be cleanly manipulated, we should not pretend the measured mediator is automatically causal. Use sensitivity analysis or design-based evidence to strengthen the mediation claim.

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